

QUANTUM CONSTANT

At the level of individual atoms and subatomic particles, electrically charged matter emits and absorbs electromagnetic radiation as small packets called photons. These tiny objects are “quanta” (discrete, self-contained bundles) of electromagnetic energy. The amount of energy a photon contains is determined by a simple calculation involving its frequency, the speed of light, and a number known as Planck’s constant.

PHOTON’S ENERGY
The energy carried by a single photon is greater for higher frequencies and shorter wavelengths of light.

FREQUENCY
The frequency of a photon is the number of waves that would pass a point in a single second if the wave was continuous—as the photon’s wavelength shrinks, its frequency increases.

$$E = h \nu$$

PLANCK’S CONSTANT
This fundamental constant of quantum physics defines the relationship between a photon’s energy and its frequency or wavelength. It means that energy is only delivered in discrete units.

$$h = 6.62607015 \times 10^{-34}$$

PLANCK’S CONSTANT

JOULE
SECONDS